1 INSTRUCTION FOR REVIEWERS

We're aware this a very rough draft that needs polishing!

Things we will sort out:

1. The results might be quite confusing at the moment. Especially as we haven't decided on a single 'metric' to measure performance, and we keep changing what's reported.
2. Figures will also be simplified (especially fig 4/5), but it will still be good to know which ones are most confusing.
3. More detailed methods, stats, and further analyses to go in Appendix are incomplete.

We would appreciate feedback on:

1. Whether the overall point and approach makes sense. Could you roughly understand how you'd replicate the work?
2. Framing. Currently we're undecided whether to frame our paper around scalable oversight.
   For - this is that it simplifies the story, and gives more context as to why we ran the particular experiments and analyses. In particular, the 'core' method (of using historic information to predict future models' performance on harder tasks) seems unnecessary unless we're trying to establish grounding for developing an oversight protocol.
   Against - we don't actually show any protocol that would work for scalable oversight, and "scalable oversight" is a loaded term that usually implies a broader set of assumptions and desiderata that "capability estimation beyond benchmark saturation" may not fulfill. the current setting may be defensible even without this framing that people may overindex on. Existing analyses can instead be framed directly as tests of whether transcript-based capability estimation remains reliable under the distribution shifts that matter for extending evaluation lifetime, including shifts across time, model paradigms, capability levels, and task domains etc.
3. Overall feedback on current results figures; which ones make sense as is, which ones require more thought/analysis to understand but are understandable regardless, which ones take too much time to try to decipher. Specific suggestions on how to improve/clarify figures further would be greatly appreciated.

EVALUATING MODEL CAPABILITY AFTER BENCHMARK SATURATION

Anonymous authors
Paper under double-blind review

ABSTRACT

As artificial intelligence systems evolve and integrate into our society, it becomes increasingly important to have well-designed evaluations to provide evidence about system capabilities and the risks that follow. Developing and validating sufficiently difficult replacement benchmarks requires substantial expertise and effort, creating a gap in which model capabilities may advance faster than available evaluations can be built to measure them. Even after benchmark scores saturate, however, the underlying evaluation transcripts may still preserve differences in how models solve the same tasks. We thus study whether these transcript-level differences can be used to estimate capability beyond benchmark saturation. We retrospectively evaluate representative contemporary approaches for extracting information from model transcripts, including pairwise LLM judgments, single-transcript absolute scoring, deterministic transcript features, and semantic transcript representations. We find that transcripts from saturated benchmark items retain capability information beyond benchmark accuracy: pairwise LLM judging reduces prediction error by $6.7\times$ compared to baseline that only uses benchmark scores. We also find that capability signal is distributed throughout transcripts and can transfer predictive ability across some capability domains, but current protocols are not robust to distribution shift. In particular, we identify a systematic underestimation of model capability with the introduction of reasoning models in early 2025, showing that current capability estimation protocols can fail under the very paradigm shifts we hope to be able to predict across. Extending the lifetime of existing evaluations will thus require methods that are better designed to extract these residual capability signals more robustly.

2 INTRODUCTION

As AI systems become increasingly integrated in society, it becomes critical that we possess reliable means to monitor the progress of frontier AI capabilities. At present, we largely rely on evaluation benchmarks for this, which have been instrumental in informing policy decisions ranging from socioeconomic to biosafety or cybersecurity concerns (OpenAI, 2025; Anthropic, 2026; Flynn et al., 2025). Yet the rapid progress of AI in has led to many existing benchmarks becoming saturated, with many models score at or near perfect accuracy. This degrades the reliability benchmark scores to estimate capabilities.

One solution to this is to develop new, more difficult benchmarks (Center for AI Safety et al., 2026; Wang et al., 2024b; Kiela et al., 2021; White et al., 2025; Jain et al., 2025; Ying et al., 2024). Yet the substantial time and resources required to design new benchmarks means these efforts are outpaced by frontier AI model capabilities [CITE]. With AI capabilities continuing to accelerate toward expert human performance, it may soon become altogether unviable to produce new benchmarks of sufficient difficulty.

Yet benchmarks produce far more content than the scalar scores we typically extract from them: a full transcript capturing *how* a model solved each task. Previous work has indicated that these transcripts may contain capability-relevant information (Zheng et al., 2023; Liu et al., 2023; Kim et al., 2024; Uesato et al., 2022; Lightman et al., 2024; Lin et al., 2022; Kadavath et al., 2022; Tian et al., 2023). For example, an LLM-judge can predict the correctness of a model on a task by evaluating its intermediate reasoning steps (Zheng et al., 2023; Liu et al., 2023; Kim et al., 2024;

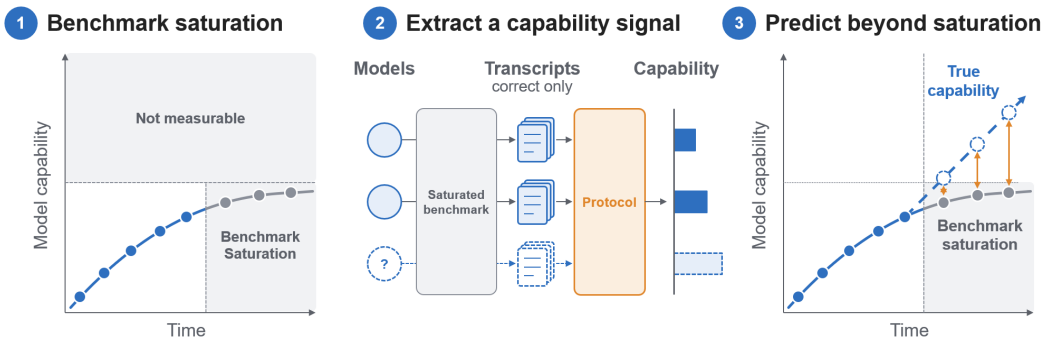


Figure 1: **Overview of our evaluation setting.** As benchmark scores saturate, we test whether transcripts from correctly solved items retain additional capability information. Using earlier models as references, transcript-based protocols predict target-model capability on harder, unsaturated items.

Uesato et al., 2022; Lightman et al., 2024). Such evidence demonstrates that benchmark transcripts contain an LLM-extractable signal that is informative about correctness.

We extend this work by asking: can we use transcripts from fully saturated benchmarks to estimate model *capabilities*? And by doing so, can we overcome the lack of signal provided by saturated benchmark scores, thus continuing to estimate capabilities beyond benchmark saturation?

We instantiate this question by assessing different methods for estimating capabilities from saturated benchmark transcripts. These methods were selected to represent contemporary approaches for extracting capability information from model transcripts, including pairwise LLM comparison (Zheng et al., 2023), single-transcript scoring (Liu et al., 2023; Kim et al., 2024), deterministic transcript features (Bitton et al., 2025), and semantic transcript representations. We then evaluate whether these approaches recover capability information once benchmark accuracy no longer reliably distinguishes between models.

Our main findings are as follows:

1. **Transcripts from correctly solved saturated items retain capability information beyond benchmark accuracy.** Pairwise LLM judging recovers the strongest signal, predicting capability on harder, unsaturated items substantially better than a benchmark accuracy-based baseline.
2. **Capability-relevant information is distributed across transcripts.** The capability signal is not concentrated in a single transcript position, while the importance of specific channels such as reasoning and tool use varies across domains.
3. **Current capability-estimation protocols are not robust to all distribution shifts.** Prediction performance notably degrades under larger shifts such as the introduction of reasoning models, motivating more generalizable methods for transcript-based capability estimation.

3 Problem Setting

We consider a setting in which model capabilities improve faster than new benchmarks can be developed and validated. When differences in benchmark scores can no longer be used to separate models of different capabilities, we consider the benchmark to be *saturated*.

Let $\mathcal{M}_{\text{ref}}$ denote a set of reference models for which capability estimates are available. Although the saturated benchmark scores no longer provide sufficient discrimination, evaluating each reference model $r \in \mathcal{M}_{\text{ref}}$ still produces a set of transcripts $\mathcal{T}_r$ that record its behavior on the benchmark. For each reference model, we assume access to these transcripts and to an estimate of its capability θ_r. We collect this information into the reference dataset $\mathcal{D}_{\text{ref}}$.

$$\mathcal{D}_{\text{ref}} = \{(\mathcal{T}_r, \theta_r) : r \in \mathcal{M}_{\text{ref}}\}.$$

We then consider a set of target models $\mathcal{M}_{\text{target}}$ whose capabilities are to be estimated. For each target model $m \in \mathcal{M}_{\text{target}}$, we observe its set of transcripts $\mathcal{T}_m$ on the same saturated benchmark, but its capability θ_m is unknown.

The problem is to estimate the capability of each target model from the reference data and the target model's transcripts. A capability estimation *protocol* $\mathcal{P}$ therefore produces

$$\mathcal{P}\left(\mathcal{D}_{\text{ref}}, \mathcal{T}_m\right) \to \hat{\theta}_m, \qquad m \in \mathcal{M}_{\text{target}}.$$

4 Experimental Methods

We instantiate our setting retrospectively using historical model releases and benchmark results. For each prediction task, reference models precede the target period. We additionally split benchmark items into an observed *saturated set* and a harder *unsaturated set*. Protocols receive only transcripts from the saturated set, while the unsaturated set is used exclusively to measure the target capability θ^*_m that the protocols attempt to predict.

4.1 Evaluation Setup

Reference and target periods. We divide model releases into consecutive semesters. For each prediction task, the *reference set* consists of models available in the preceding reference period $\mathcal{M}_{t-1}$, while the *target set* consists of models released during the subsequent target period $\mathcal{M}_t$. Our main evaluation uses three target periods: early 2025, late 2025, and early 2026, with eight reference models and eight target models in each period.

Determining the saturation boundary. For each target period, we identify the three highest-capability models available for calibration. We define an item as belonging to the *saturated set* if all three models answer it correctly. The remaining items form the *unsaturated set*.

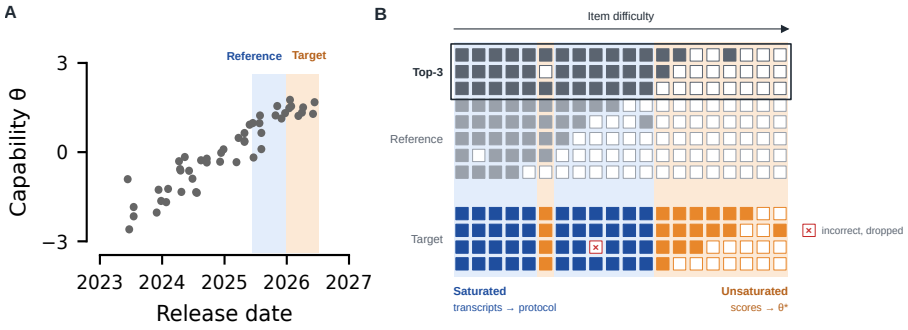


Figure 2: The model capability θ_m, fitted over the science domain for all semesters.

Prediction target. To measure the capability beyond the saturated portion of the benchmark, we fit a local IRT model to the reference and target models on the unsaturated set while keeping each item's difficulty parameter b_i fixed across semesters. We use the measured capability θ^* of the target models as the prediction target. We provide the full IRT fitting procedure in Appendix A.

Table 1: **Evaluation domains and benchmarks.** Benchmarks used in each capability domain.

Domain	Benchmarks
Science	GPQA-Diamond (Rein et al., 2024), ARC-Challenge (Clark et al., 2018), HLE-Gold [Science] (Zhai et al., 2026)
Mathematics	MMLU-Pro [Math] (Wang et al., 2024b), AIME 2024 (Jia, 2024), AIME 2025, AIME 2026, APEX 2025, APEX Shortlist (Dekoninck et al., 2026), HLE-Gold [Math]
Coding	AgentBench OS (Liu et al., 2024), SWE-Bench (Jimenez et al., 2024), Terminal-Bench 2.1 (Merrill et al., 2026)
Cybersecurity	CyBench (Zhang et al., 2025), ExploitBench (Lee & Brumley, 2026)

Transcripts. Our primary experiments are designed to extract the signal of capability contained in the transcripts of the saturated tasks, focusing on *how* models solve the tasks. The proposed protocols therefore receive only transcripts from saturated items that the evaluated model answers correctly. Unless otherwise stated, we utilize the complete model response.

Domains. We evaluate four domains: science, mathematics, coding, and cybersecurity. Each domain combines benchmarks spanning a range of item difficulties, allowing us to distinguish between saturated and unsaturated items. Table 1 summarizes the benchmarks used in each domain.

Model selection. We primarily use open-weight models because the main evaluation setting utilizes reasoning traces, which are generally not exposed by closed-source APIs, making controlled comparison difficult. We therefore restrict the main experiments to models for which complete transcript access is available, and evaluate reasoning closed-source models separately in Section 5.3. The full model list is provided in Appendix B.1.

Metrics. For a protocol $\mathcal{P}$ and target model m, we measure prediction error using squared error: $\epsilon_m = \left(\hat{\theta}_m - \theta^*_m\right)^2$, where $\hat{\theta}_m$ is the predicted capability and θ^*_m is the measured target capability.

4.2 Prediction Protocols

We compare several contemporary approaches for extracting capability information from the observations available in our setting. Table 2 summarizes the protocols; full implementation details are provided in Appendix C.

Table 2: Prediction protocols evaluated. All non-baseline protocols aggregate transcript-level signals at the model level and calibrate them to target capability θ^* using the reference models.

Protocol	Signal extracted
Baseline	No transcript information
Pairwise judging	Relative capability evidence from comparisons between two transcripts
Absolute scoring	Capability-related score assigned to each transcript independently
Transcript features	Deterministic properties (e.g., transcript length and prose features)
Transcript embeddings	Fixed text-embedding representation of each transcript

Baseline prediction. The baseline represents the information available from benchmark accuracy once the observed region is saturated. Since all protocols observe only transcripts from items that the evaluated model answers correctly, accuracy on these items cannot distinguish the target models and thus predicts the same capability for every target model, equal to the mean measured capability of the reference models $\bar{\theta}_{\text{ref}}$.

5 Results

We evaluate transcript-based capability estimation after benchmark saturation through three questions: **(1)** Do correctly solved transcripts from saturated benchmarks contain capability signal beyond the benchmark score? **(2)** Where in the transcript is this signal located? **(3)** When does this signal generalize across time, models, and domains?

5.1 Saturated benchmark transcripts contain capability information

We first ask whether transcripts from saturated benchmark items contain capability information. For each target period, protocols receive reference models and transcripts from saturated items that the evaluated models answer correctly; they must then predict capability measured on the harder, unsaturated items. Figure 3 compares the prediction performance of the four transcript-based protocols across our evaluation settings.

Pairwise transcript judging produced the lowest prediction error of the tested protocols, with an average squared error of [TODO: ...], compared with [TODO: ...] for the constant baseline. This improvement was observed across all four capability domains, although its magnitude differed substantially between them: [TODO: insert squared errors for science, mathematics, coding, and cybersecurity]. The remaining transcript-based approaches were less reliable. Transcript embeddings achieved lower error than the baseline in [TODO: settings/domains], but not consistently across

the full evaluation. Single-transcript absolute scoring and deterministic transcript features generally produced errors comparable to or larger than the baseline. [TODO: Insert exact squared-error summary after recomputation.] A supplementary likelihood-ratio analysis found that prediction error differed systematically across protocols ([TODO: $\chi^2(df) = X, p = Y$]).

If capability prediction were driven mainly by simple properties such as response length or surface-level writing style, these feature-based predictors should recover similar information. Instead, pairwise comparison performs substantially better, suggesting that the useful signal includes richer differences in model behavior that are accessible to an LLM judge. However, the large differences between protocols also show that this information may not always be straightforward to extract.

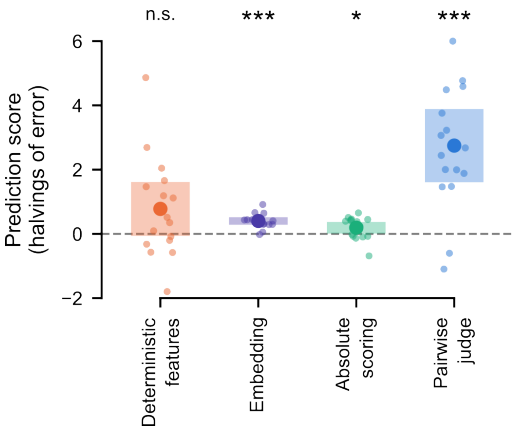


Figure 3: **Saturated transcripts contain capability-predicting signal.** Mean prediction scores for four representative protocols. Positive scores indicate higher accuracy than the baseline. *: $p < .05$, ***: $p < .001$, one-sample t-test on prediction scores for $N = 16$ target models. Shading represents 95% CI. [TODO: please rotate x axis labels, we're wasting precious space :(]

5.2 Capability signals are distributed throughout transcripts

Having established that saturated benchmark transcripts contain capability signal, we next ask *where* this signal lies. This is practically important because full reasoning traces may be unavailable for closed-source models and may be too long to process efficiently in reasoning or agentic settings. We focus the main analysis on pairwise judging, the strongest-performing protocol in Section 5.1, and report corresponding analyses for the remaining protocols in Appendix **??**.

5.2.1 Capability information persists across transcript positions

To examine how capability-predicting information is distributed across transcript sections, we divided every transcript into thirds: a start, a middle, and end. We then tested whether a section was necessary for capability prediction by removing it from a transcript and measuring prediction degradation of pairwise judging.

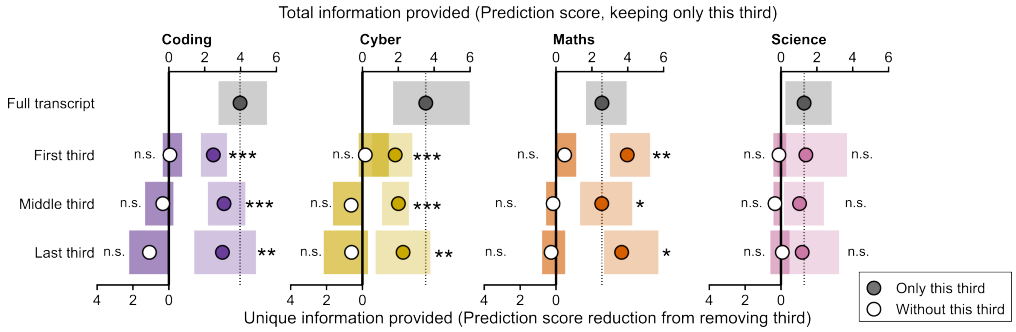


Figure 4: **Capability information persists across transcript positions.** Squared prediction error when pairwise judging receives only the beginning, middle, or end of each transcript, and when each corresponding section is removed from the complete transcript. Lower values indicate more accurate capability estimates. [TODO: Replace existing prediction-score axes and recompute uncertainty intervals using squared error.] */**/***: $p < .05/< .01/< .001$, bootstrapped 95% CI over prediction targets

Across all domains, we found no indication that any section was necessary: removing sections did not significantly reduce capability prediction scores. With the exception of the science domain, keeping *any* third of the transcript still yielded above-chance performance that was indistinguishable

from the full transcript ([TODO: $\chi^2(df) = X$, $p = Y$ or some other stats details]). These results suggest that capability estimation may remain possible even when the judge cannot observe the complete transcript, as information useful to the pairwise judge appears repeatedly throughout the trajectory [TODO: we also show if a segment is sufficient for predictions, by only giving judges access to that section. in appendix]

5.2.2 Capability signals can be concentrated in specific channels

We next tested whether capability-relevant signals are concentrated in specific channels, such as the model's messages, reasoning, or tool calls. Across the four domains, messages account for only 3–9% of transcripts, while reasoning dominates mathematics and science (95% and 91%) and tool calls dominate cybersecurity and coding (66% and 68%). We therefore expected the dominant channel in each domain to carry the most capability-relevant information.

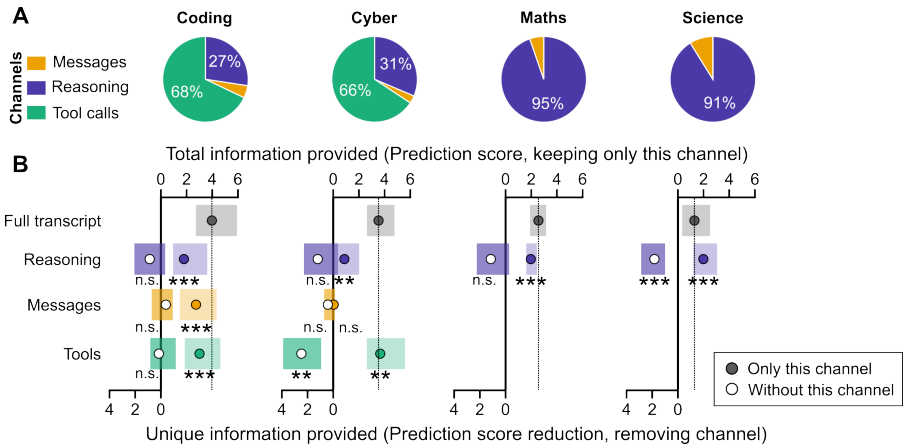


Figure 5: **Capability signals are differentially distributed across channels in different domains A.** Proportion of transcripts corresponding to different channels in each domain. **B.** Prediction score when keeping only individual channels (**top**-axis; dark circles on light box) and reduction in prediction score after removing that channel, relative to the full transcript (**bottom**-axis, white circle on dark box) across each domain. [TODO: Replace prediction-score axes and recompute uncertainty intervals using squared error.]

Removing any single channel produces little change in coding or mathematics, whereas removing reasoning from science increases squared error from [TODO: . . .] to [TODO: . . .], and removing tool interactions from cybersecurity increases it from [TODO: . . .] to [TODO: . . .]. [TODO: LLR statistics if useful.] Thus, capability signal depends on both channel content and domain rather than transcript share alone. In particular, restricted access to reasoning traces may reduce the capability signal recoverable from some domains.

5.3 Generalisation of capability predictions

Thus far, our results show that capability-relevant information is present throughout saturated benchmark transcripts, and that this can be extracted by several basic protocols. Yet to have confidence that a protocol will work outside the specific cases we test, we propose several generalisations that protocols must be robust to.

Prediction reliability varies across historical periods. We first ask whether protocol performance is stable across time. We retrospectively repeat the prediction across time. Science and mathematics are back-tested to early 2024, while coding and cybersecurity begin in early 2025 because earlier models provide insufficient coverage of the respective agentic benchmarks.

Prediction error varies substantially across historical periods for all four protocols. Pairwise judging generally achieves the lowest error, but its performance deteriorates sharply in particular settings, including mathematics in early 2025 and science in [TODO: verify period]. The other protocols show similarly large variation across periods. [TODO: Insert SE ranges and representative values.]

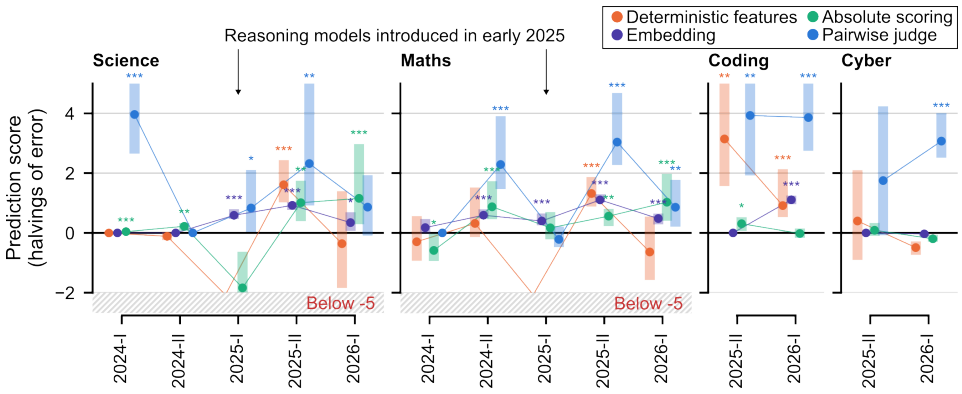


Figure 6: **Prediction error varies substantially across historical target periods.** Squared prediction error for each protocol when calibrated on the preceding reference population and evaluated on models from each target period. [TODO: Replace prediction-score axis and statistical annotations with squared-error results.]

Generalisation across paradigm shifts. The largest mathematics failure occurs in early 2025 (Figure **??**), coinciding with the introduction of reasoning models and a sharp increase in mathematical capability. We therefore use this period to test whether the error is concentrated in this new model class.

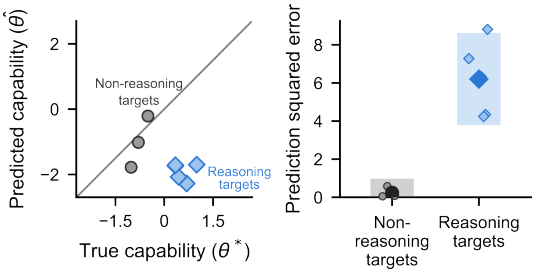


Figure 7: **Pairwise judging systematically underestimates reasoning models in early-2025 mathematics. Left:** Measured capability θ^*_m against predicted capability $\hat{\theta}_m$ for early-2025 mathematics targets. **Right:** Squared prediction error for reasoning and non-reasoning target models. [TODO: Update numerical values and uncertainty intervals.]

Pairwise judging yields squared error of [TODO: . . .] for non-reasoning targets and [TODO: . . .] for reasoning targets. A likelihood-ratio analysis finds [TODO: effect] of target-model type on prediction error ([TODO: $\chi^2(df) = X,\ p = Y$]). Signed residuals, $\hat{\theta}_m - \theta^*_m$, show systematic underestimation of reasoning models, indicating that the relationship between behavior on saturated tasks and capability on harder items shifts for this model class. Calibration learned from earlier models therefore fails to extrapolate across this transition.

Open-weight to closed-source models transfer. Our experiments on the effects of reasoning traces show that access to complete transcripts allow for stronger predictive performance. We extend this analysis for closed-source models, whose architectures, training procedures, and exposed outputs can differ from the open-weight models used for calibration regardless of reasoning trace access. [TODO: analyze]

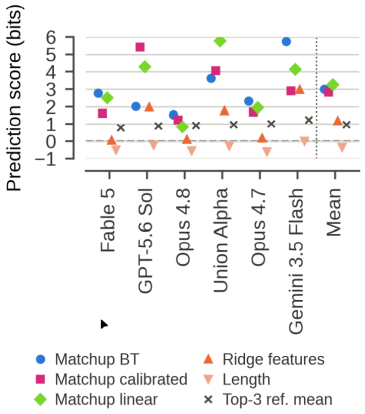


Figure 8: **Transfer from open-weight reference models to closed-source target models.** Prediction performance when the target population matches the open-weight reference population versus when the same protocol is applied to closed-source targets.

Generalisation across domains. Finally, we explored whether capability predictions derived from transcripts from one domain could predict capabilities in another domain. To do so, we took transcripts from one domain, and calibrated predictions for pairwise judging to the reference models in another domain (see Section 4: Methods). Surprisngly, we found that capabilities in some domains were *better* predicted using transcripts from other domains (Figure 9). For example, transcripts from science or coding benchmarks could predict maths capabilities better than within-domain transcripts. The reverse

was not true however: maths transcripts performed poorly in predicting performance in coding, amplifying errors by 17.6x [TODO: proper analyses]. These patterns may indicate that the limitation of capability prediction in maths may be due primarily to poor signals in the transcripts

6 Related Work

Benchmark saturation and evaluation design. Previous works on benchmark report the impact of score saturation and the downstream limit to discrimination amongst strong models (Bowman & Dahl, 2021; Vania et al., 2021; Akhtar et al., 2026). Many works address and attempt to resolve this by introducing new challenging versions of existing benchmarks (Hendrycks et al., 2021; Suzgun et al., 2023; Rein et al., 2024; Wang et al., 2024b; Center for AI Safety et al., 2026). Another line of works also explore the idea of dynamic evaluation which instead uses adversarial data collection or some continuously updated test set to maintain difficulty over time (Kiela et al., 2021; White et al., 2025; Jain et al., 2025), but costs for running these methods are much higher than alternatives.

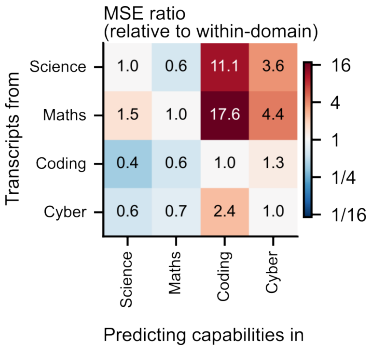


Figure 9: **Pairwise judging predictions can transfer across some domains.**

Information from model responses. Previous works on transcript-analysis extracts signals beyond answer correctness through pairwise comparison and rubric-based scoring (Zheng et al., 2023; Liu et al., 2023; Kim et al., 2024), process-level reasoning assessment (Uesato et al., 2022; Lightman et al., 2024), and confidence-based prediction of correctness (Lin et al., 2022; Kadavath et al., 2022; Tian et al., 2023). These approaches suggest that model transcripts are a promising source of information. However, LLM judges have been reported to exhibit systematic response-length and position biases (Zheng et al., 2023; Wang et al., 2024a; Dubois et al., 2024), suggesting signals can be sensitive to factors unrelated to actual capability.

Capability estimation and prediction. Previous works on Item Response Theory (IRT) uses it for latent ability estimation, item calibration, and test-set comparison (Lalor et al., 2016; 2019; Rodriguez et al., 2021; Vania et al., 2021; Zhou et al., 2026). By accounting for differences in item difficulty, IRT can be used to provide a more structured capability estimate than accuracy. Additional work on adaptive testing and representative item selection focus on a different direction of reducing evaluation cost by identifying informative subsets of tasks (Zhuang et al., 2025; Maia Polo et al., 2024; Fonseca Rivera et al., 2026), although estimation reliability depends on the number and distribution of evaluated models (Jiang et al., 2026). Finally, forecasting work predicts benchmark performance from signals on other tasks (Owen, 2024; Ruan et al., 2024).

7 Discussion

Our results suggest that even when benchmark accuracy can no longer be used to compare model capability, transcripts still retain information that can be extracted. Among the protocols we study, pairwise LLM judging recovers the strongest signal, while simpler approaches based on individual transcript scores, embeddings, or surface-level features are notably less reliable.

However, demonstrating that capability information is present in transcripts is only the first step. A useful capability-estimation protocol must also extract that information reliably when the relationship between transcript behavior and capability changes. Our results show that current protocols do not yet meet this requirement. Prediction performance varies substantially across time, and the transition to reasoning models in early 2025 provides a clear failure case in which judging-based protocols systematically underestimate the capabilities of the new model class. Future work should therefore focus not only on extracting *more* signal from transcripts, but on designing protocols that remain robust under distribution shift.

AI use statement

(This section is **required** and does not count toward the page limit.)

In this work, we used generative AI tools for [tasks with required disclosure]. We have not used generative AI tools for [other tasks with required disclosure], and [the rest of the required disclosure tasks] are not applicable to this work. Additionally, we used generative AI tools for [tasks with recommended disclosure]. We have reviewed all AI-assisted work. [Elaborate. For example, "we checked LLM-generated research ideas for potential plagiarism through a manual literature survey", "LLM-generated code was verified and tested for correctness by 2 authors", etc.]. We take responsibility for the final content of this work, including text, claims or artifacts produced with the aid of generative AI.

Reproducibility statement

(This section is **recommended** and does not count toward the page limit.)

It is important that the work published in ICLR is reproducible. Authors are strongly encouraged to include a paragraph-long Reproducibility Statement at the end of the main text (before references) to discuss the efforts that have been made to ensure reproducibility. This paragraph should not itself describe details needed for reproducing the results, but rather reference the parts of the main paper, appendix, and supplemental materials that will help with reproducibility. For example, for novel models or algorithms, a link to an anonymous downloadable source code can be submitted as supplementary materials; for theoretical results, clear explanations of any assumptions and a complete proof of the claims can be included in the appendix; for any datasets used in the experiments, a complete description of the data processing steps can be provided in the supplementary materials. Each of the above are examples of things that can be referenced in the reproducibility statement. [TODO: mention anonymized github which includes data collection and processing of existing data; anonymized data release; implementation details of each protocol in appendix]

References

Mubashara Akhtar, Anka Reuel, Prajna Soni, Sanchit Ahuja, Pawan Sasanka Ammanamanchi, Ruchit Rawal, Vilém Zouhar, Srishti Yadav, Chenxi Whitehouse, Dayeon Ki, Jennifer Mickel, Leshem Choshen, Marek Šuppa, Jan Batzner, Jenny Chim, Jeba Sania, Yanan Long, Hossein A. Rahmani, Christina Knight, Yiyang Nan, Jyoutir Raj, Yu Fan, Shubham Singh, Subramanyam Sahoo, Eliya Habba, Usman Gohar, Siddhesh Pawar, Robert Scholz, Arjun Subramonian, Jingwei Ni, Mykel Kochenderfer, Sanmi Koyejo, Mrinmaya Sachan, Stella Biderman, Zeerak Talat, Avijit Ghosh, and Irene Solaiman. When AI benchmarks plateau: A systematic study of benchmark saturation. In *Proceedings of the 43rd International Conference on Machine Learning*, 2026. URL https://arxiv.org/abs/2602.16763.

Anthropic. Anthropic's responsible scaling policy. https://www.anthropic.com/responsible-scaling-policy, 2026. URL https://www.anthropic.com/responsible-scaling-policy. Version 3.4, effective July 8, 2026. Accessed September 17, 2026.

Yehonatan Bitton, Elad Bitton, and Shai Nisan. Detecting stylistic fingerprints of large language models. *arXiv preprint arXiv:2503.01659*, 2025. URL https://arxiv.org/abs/2503.01659.

Samuel R. Bowman and George E. Dahl. What will it take to fix benchmarking in natural language understanding? In *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pp. 4843–4855, Online, June 2021. Association for Computational Linguistics. doi: 10.18653/v1/2021.naacl-main.385. URL https://aclanthology.org/2021.naacl-main.385/.

Center for AI Safety, Scale AI, and HLE Contributors Consortium. A benchmark of expert-level academic questions to assess AI capabilities. *Nature*, 649(8099):1139–1146, January 2026. doi: 10.1038/s41586-025-09962-4. URL https://www.nature.com/articles/s41586-025-09962-4.

Peter Clark, Isaac Cowhey, Oren Etzioni, Tushar Khot, Ashish Sabharwal, Carissa Schoenick, and Oyvind Tafjord. Think you have solved question answering? try ARC, the AI2 reasoning chal-

lenge. *arXiv preprint arXiv:1803.05457*, 2018. URL https://arxiv.org/abs/1803.05457.

Jasper Dekoninck, Nikola Jovanović, Tim Gehrunger, Kári Rögnvaldsson, Ivo Petrov, Chenhao Sun, and Martin Vechev. Beyond benchmarks: MathArena as an evaluation platform for mathematics with LLMs. *arXiv preprint arXiv:2605.00674*, 2026. URL https://arxiv.org/abs/2605.00674.

Yann Dubois, Percy Liang, and Tatsunori Hashimoto. Length-controlled alpacaeval: A simple debiasing of automatic evaluators. In *First Conference on Language Modeling*, 2024. URL https://openreview.net/forum?id=CybBmzWBX0.

Four Flynn, Helen King, and Anca Dragan. Strengthening our Frontier Safety Framework. Google DeepMind blog, September 2025. URL https://deepmind.google/blog/strengthening-our-frontier-safety-framework/. Published September 22, 2025. Updated April 17, 2026 (Frontier Safety Framework v3.1).

Joshua Fonseca Rivera, Neil Shah, David Demitri Africa, and Konstantinos Voudouris. Item Response Theory for AI safety. arXiv preprint arXiv:2608.05086, 2026. URL https://arxiv.org/abs/2608.05086.

Dan Hendrycks, Collin Burns, Saurav Kadavath, Akul Arora, Steven Basart, Eric Tang, Dawn Song, and Jacob Steinhardt. Measuring mathematical problem solving with the MATH dataset. In *Proceedings of the Neural Information Processing Systems Track on Datasets and Benchmarks*, volume 1, 2021. URL https://datasets-benchmarks-proceedings.neurips.cc/paper/2021/file/be83ab3ecd0db773eb2dc1b0a17836a1-Paper-round2.pdf.

Naman Jain, King Han, Alex Gu, Wen-Ding Li, Fanjia Yan, Tianjun Zhang, Sida Wang, Armando Solar-Lezama, Koushik Sen, and Ion Stoica. LiveCodeBench: Holistic and contamination free evaluation of large language models for code. In *The Thirteenth International Conference on Learning Representations*, 2025. URL https://openreview.net/forum?id=chfJJYC3iL.

Maxwell Jia. AIME 2024 dataset. Hugging Face dataset, December 2024. URL https://huggingface.co/datasets/Maxwell-Jia/AIME_2024.

Han Jiang, Sunbeom Kwon, Jinwen Luo, Ziang Xiao, and Susu Zhang. Can we trust Item Response Theory for AI evaluation? arXiv preprint arXiv:2607.15190, 2026. URL https://arxiv.org/abs/2607.15190.

Carlos E. Jimenez, John Yang, Alexander Wettig, Shunyu Yao, Kexin Pei, Ofir Press, and Karthik R. Narasimhan. SWE-bench: Can language models resolve real-world github issues? In *The Twelfth International Conference on Learning Representations*, 2024. URL https://openreview.net/forum?id=VTF8yNQM66.

Saurav Kadavath, Tom Conerly, Amanda Askell, Tom Henighan, Dawn Drain, Ethan Perez, Nicholas Schiefer, Zac Hatfield-Dodds, Nova DasSarma, Eli Tran-Johnson, Scott Johnston, Sheer El-Showk, Andy Jones, Nelson Elhage, Tristan Hume, Anna Chen, Yuntao Bai, Sam Bowman, Stanislav Fort, Deep Ganguli, Danny Hernandez, Josh Jacobson, Jackson Kernion, Shauna Kravec, Liane Lovitt, Kamal Ndousse, Catherine Olsson, Sam Ringer, Dario Amodei, Tom Brown, Jack Clark, Nicholas Joseph, Ben Mann, Sam McCandlish, Chris Olah, and Jared Kaplan. Language models (mostly) know what they know. arXiv preprint arXiv:2207.05221, 2022. URL https://arxiv.org/abs/2207.05221.

Douwe Kiela, Max Bartolo, Yixin Nie, Divyansh Kaushik, Atticus Geiger, Zhengxuan Wu, Bertie Vidgen, Grusha Prasad, Amanpreet Singh, Pratik Ringshia, Zhiyi Ma, Tristan Thrush, Sebastian Riedel, Zeerak Waseem, Pontus Stenetorp, Robin Jia, Mohit Bansal, Christopher Potts, and Adina Williams. Dynabench: Rethinking benchmarking in NLP. In *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pp. 4110–4124, Online, June 2021. Association for Computational Linguistics. doi: 10.18653/v1/2021.naacl-main.324. URL https://aclanthology.org/2021.naacl-main.324/.

Seungone Kim, Juyoung Suk, Shayne Longpre, Bill Yuchen Lin, Jamin Shin, Sean Welleck, Graham Neubig, Moontae Lee, Kyungjae Lee, and Minjoon Seo. Prometheus 2: An open source language model specialized in evaluating other language models. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pp. 4334–4353, Miami, Florida, USA, November 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.emnlp-main.248. URL https://aclanthology.org/2024.emnlp-main.248/.

John P. Lalor, Hao Wu, and Hong Yu. Building an evaluation scale using Item Response Theory. In *Proceedings of the 2016 Conference on Empirical Methods in Natural Language Processing*, pp. 648–657, Austin, Texas, November 2016. Association for Computational Linguistics. doi: 10.18653/v1/D16-1062. URL https://aclanthology.org/D16-1062/.

John P. Lalor, Hao Wu, and Hong Yu. Learning latent parameters without human response patterns: Item Response Theory with artificial crowds. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, pp. 4249–4259, Hong Kong, China, November 2019. Association for Computational Linguistics. doi: 10.18653/v1/D19-1434. URL https://aclanthology.org/D19-1434/.

Seunghyun Lee and David Brumley. ExploitBench: A capability ladder benchmark for LLM cybersecurity agents. arXiv preprint arXiv:2605.14153, 2026. URL https://arxiv.org/abs/2605.14153.

Hunter Lightman, Vineet Kosaraju, Yura Burda, Harri Edwards, Bowen Baker, Teddy Lee, Jan Leike, John Schulman, Ilya Sutskever, and Karl Cobbe. Let's verify step by step. In *The Twelfth International Conference on Learning Representations*, 2024. URL https://openreview.net/forum?id=v8L0pN6EOi.

Stephanie Lin, Jacob Hilton, and Owain Evans. Teaching models to express their uncertainty in words. *Transactions on Machine Learning Research*, 2022. ISSN 2835-8856. URL https://openreview.net/forum?id=8s8K2UZGTZ.

Xiao Liu, Hao Yu, Hanchen Zhang, Yifan Xu, Xuanyu Lei, Hanyu Lai, Yu Gu, Hangliang Ding, Kaiwen Men, Kejuan Yang, Shudan Zhang, Xiang Deng, Aohan Zeng, Zhengxiao Du, Chenhui Zhang, Sheng Shen, Tianjun Zhang, Yu Su, Huan Sun, Minlie Huang, Yuxiao Dong, and Jie Tang. AgentBench: Evaluating LLMs as agents. In *The Twelfth International Conference on Learning Representations*, 2024. URL https://arxiv.org/abs/2308.03688.

Yang Liu, Dan Iter, Yichong Xu, Shuohang Wang, Ruochen Xu, and Chenguang Zhu. G-Eval: NLG evaluation using GPT-4 with better human alignment. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pp. 2511–2522, Singapore, December 2023. Association for Computational Linguistics. doi: 10.18653/v1/2023.emnlp-main.153. URL https://aclanthology.org/2023.emnlp-main.153/.

Felipe Maia Polo, Lucas Weber, Leshem Choshen, Yuekai Sun, Gongjun Xu, and Mikhail Yurochkin. tinyBenchmarks: Evaluating LLMs with fewer examples. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pp. 34303–34326. PMLR, 2024. URL https://proceedings.mlr.press/v235/maia-polo24a.html.

Mike A. Merrill, Alexander G. Shaw, Nicholas Carlini, Boxuan Li, Harsh Raj, Ivan Bercovich, Lin Shi, Jeong Yeon Shin, Thomas Walshe, E. Kelly Buchanan, Junhong Shen, Guanghao Ye, Haowei Lin, Jason Poulos, Maoyu Wang, Marianna Nezhurina, Jenia Jitsev, Di Lu, Orfeas Menis Mastromichalakis, Zhiwei Xu, et al. Terminal-Bench: Benchmarking agents on hard, realistic tasks in command line interfaces. arXiv preprint arXiv:2601.11868, 2026. URL https://arxiv.org/abs/2601.11868.

OpenAI. Our updated Preparedness Framework. https://openai.com/index/updating-our-preparedness-framework/, April 2025. URL https://openai.com/index/updating-our-preparedness-framework/. Published April 15, 2025.

David Owen. How predictable is language model benchmark performance? arXiv preprint arXiv:2401.04757, 2024. URL https://arxiv.org/abs/2401.04757.

David Rein, Betty Li Hou, Asa Cooper Stickland, Jackson Petty, Richard Yuanzhe Pang, Julien Dirani, Julian Michael, and Samuel R. Bowman. GPQA: A graduate-level google-proof Q&A benchmark. In *First Conference on Language Modeling*, 2024. URL https://openreview.net/forum?id=Ti67584b98.

Pedro Rodriguez, Joe Barrow, Alexander Hoyle, John P. Lalor, Robin Jia, and Jordan Boyd-Graber. Evaluation examples are not equally informative: How should that change NLP leaderboards? In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*, pp. 4486–4503, Online, August 2021. Association for Computational Linguistics. doi: 10.18653/v1/2021.acl-long.346. URL https://aclanthology.org/2021.acl-long.346/.

Yangjun Ruan, Chris J. Maddison, and Tatsunori Hashimoto. Observational scaling laws and the predictability of language model performance. In *Advances in Neural Information Processing Systems*, volume 37, 2024. doi: 10.52202/079017-0506. URL https://proceedings.neurips.cc/paper_files/paper/2024/hash/1cded4f97cf5f01a284c574110b7e3b9-Abstract-Conference.html.

Mirac Suzgun, Nathan Scales, Nathanael Schärli, Sebastian Gehrmann, Yi Tay, Hyung Won Chung, Aakanksha Chowdhery, Quoc Le, Ed Chi, Denny Zhou, and Jason Wei. Challenging BIG-Bench tasks and whether chain-of-thought can solve them. In *Findings of the Association for Computational Linguistics: ACL 2023*, pp. 13003–13051, Toronto, Canada, July 2023. Association for Computational Linguistics. doi: 10.18653/v1/2023.findings-acl.824. URL https://aclanthology.org/2023.findings-acl.824/.

Katherine Tian, Eric Mitchell, Allan Zhou, Archit Sharma, Rafael Rafailov, Huaxiu Yao, Chelsea Finn, and Christopher D. Manning. Just ask for calibration: Strategies for eliciting calibrated confidence scores from language models fine-tuned with human feedback. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pp. 5433–5442, Singapore, December 2023. Association for Computational Linguistics. doi: 10.18653/v1/2023.emnlp-main.330. URL https://aclanthology.org/2023.emnlp-main.330/.

Jonathan Uesato, Nate Kushman, Ramana Kumar, Francis Song, Noah Siegel, Lisa Wang, Antonia Creswell, Geoffrey Irving, and Irina Higgins. Solving math word problems with process- and outcome-based feedback. arXiv preprint arXiv:2211.14275, 2022. URL https://arxiv.org/abs/2211.14275.

Clara Vania, Phu Mon Htut, William Huang, Dhara Mungra, Richard Yuanzhe Pang, Jason Phang, Haokun Liu, Kyunghyun Cho, and Samuel R. Bowman. Comparing test sets with Item Response Theory. In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*, pp. 1141–1158, Online, August 2021. Association for Computational Linguistics. doi: 10.18653/v1/2021.acl-long.92. URL https://aclanthology.org/2021.acl-long.92/.

Peiyi Wang, Lei Li, Liang Chen, Zefan Cai, Dawei Zhu, Binghuai Lin, Yunbo Cao, Lingpeng Kong, Qi Liu, Tianyu Liu, and Zhifang Sui. Large language models are not fair evaluators. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 9440–9450, Bangkok, Thailand, August 2024a. Association for Computational Linguistics. doi: 10.18653/v1/2024.acl-long.511. URL https://aclanthology.org/2024.acl-long.511/.

Yubo Wang, Xueguang Ma, Ge Zhang, Yuansheng Ni, Abhranil Chandra, Shiguang Guo, Weiming Ren, Aaran Arulraj, Xuan He, Ziyan Jiang, Tianle Li, Max Ku, Kai Wang, Alex Zhuang, Rongqi Fan, Xiang Yue, and Wenhu Chen. MMLU-Pro: A more robust and challenging multi-task language understanding benchmark. In *Advances in Neural Information Processing Systems, Datasets and Benchmarks Track*, volume 37, 2024b. URL https://arxiv.org/abs/2406.01574.

Colin White, Samuel Dooley, Manley Roberts, Arka Pal, Benjamin Feuer, Siddhartha Jain, Ravid Shwartz-Ziv, Neel Jain, Khalid Saifullah, Sreemanti Dey, Shubh-Agrawal, Sandeep Singh Sandha, Siddartha Naidu, Chinmay Hegde, Yann LeCun, Tom Goldstein, Willie Neiswanger,

and Micah Goldblum. LiveBench: A challenging, contamination-limited LLM benchmark. In *The Thirteenth International Conference on Learning Representations*, 2025. URL https://openreview.net/forum?id=sKYHBTAxVa.

Jiahao Ying, Yixin Cao, Yushi Bai, Qianru Sun, Bo Wang, Wei Tang, Zhaojun Ding, Yizhe Yang, Xuanjing Huang, and Shuicheng Yan. Automating dataset updates towards reliable and timely evaluation of large language models. In *Advances in Neural Information Processing Systems, Datasets and Benchmarks Track*, volume 37, pp. 17106–17132, 2024. URL https://proceedings.neurips.cc/paper_files/paper/2024/hash/1e89c12621c0315373f20f0aeabe5dbe-Abstract-Datasets_and_Benchmarks_Track.html.

Weiqi Zhai, Zhihai Wang, Jinghang Wang, Boyu Yang, Xiaogang Li, Xiang Xu, Bohan Wang, Peng Wang, Xingzhe Wu, Anfeng Li, Qiyuan Feng, Yuhao Zhou, Shoulin Han, Wenjie Luo, Yiyuan Li, Yaxuan Wang, Ruixian Luo, Guojie Lin, Peiyao Xiao, Chengliang Xu, Ben Wang, Zeyu Wang, Zichao Chen, Jianan Ye, Yijie Hu, Jialong Chen, Zongwen Shen, Yuliang Xu, An Yang, Bowen Yu, Dayiheng Liu, Junyang Lin, Hu Wei, Que Shen, and Bing Zhao. Hle-verified: A systematic verification and structured revision of humanity's last exam, 2026. URL https://arxiv.org/abs/2602.13964.

Andy K Zhang, Neil Perry, Riya Dulepet, Joey Ji, Celeste Menders, Justin W Lin, Eliot Jones, Gashon Hussein, Samantha Liu, Donovan Julian Jasper, Pura Peetathawatchai, Ari Glenn, Vikram Sivashankar, Daniel Zamoshchin, Leo Glikbarg, Derek Askaryar, Haoxiang Yang, Aolin Zhang, Rishi Alluri, Nathan Tran, Rinnara Sangpisit, Kenny O Oseleononmen, Dan Boneh, Daniel E. Ho, and Percy Liang. Cybench: A framework for evaluating cybersecurity capabilities and risks of language models. In *The Thirteenth International Conference on Learning Representations*, 2025. URL https://openreview.net/forum?id=tc90LV0yRL.

Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric P. Xing, Hao Zhang, Joseph E. Gonzalez, and Ion Stoica. Judging LLM-as-a-judge with MT-Bench and Chatbot Arena. In *Advances in Neural Information Processing Systems, Datasets and Benchmarks Track*, volume 36, pp. 46595–46623, 2023. doi: 10.52202/075280-2020. URL https://proceedings.neurips.cc/paper_files/paper/2023/hash/91f18a1287b398d378ef22505bf41832-Abstract-Datasets_and_Benchmarks.html.

Hongli Zhou, Hui Huang, Ziqing Zhao, Lvyuan Han, Huicheng Wang, Kehai Chen, Muyun Yang, Wei Bao, Jian Dong, Bing Xu, Conghui Zhu, Hailong Cao, and Tiejun Zhao. Lost in benchmarks? rethinking large language model benchmarking with Item Response Theory. *Proceedings of the AAAI Conference on Artificial Intelligence*, 40(41):35085–35093, March 2026. doi: 10.1609/aaai.v40i41.40814. URL https://ojs.aaai.org/index.php/AAAI/article/view/40814.

Yan Zhuang, Qi Liu, Zachary A. Pardos, Patrick C. Kyllonen, Jiyun Zu, Zhenya Huang, Shijin Wang, and Enhong Chen. Position: AI evaluation should learn from how we test humans. In *Proceedings of the 42nd International Conference on Machine Learning*, 2025. URL https://arxiv.org/abs/2306.10512.

A Full Methodology

B Additional Experimental Configuration

This appendix provides additional details on the models and inference configurations used in our retrospective evaluations. Our main experiments are organized into six-month evaluation periods. For each period, we select models released by that point for which we can obtain the response transcripts required by our transcript-based protocols. We additionally construct a separate sequence of frontier open-weight models for the historical forecasting baseline.

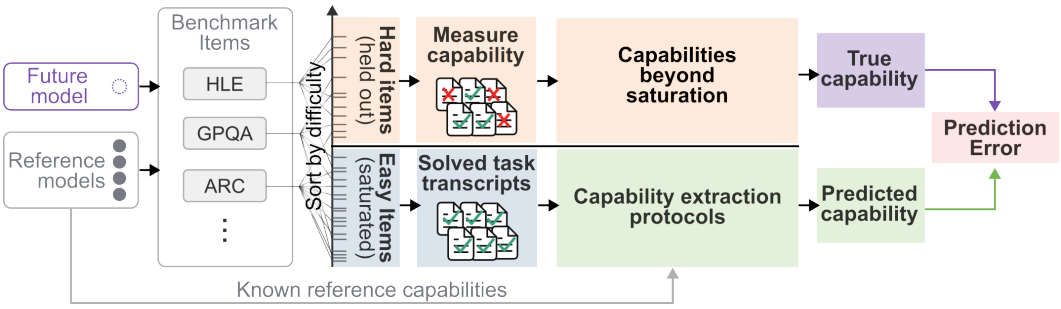


Figure 10: **Detailed methodology** [TODO: fill tis]

B.1 Evaluated Models

Table 3 lists the models included in each evaluation period. We aim to cover a range of capability levels within each period rather than selecting only the strongest available models, since the reference-based prediction protocols require variation in capability for calibration. At the same time, each period contains several models near the capability frontier at that point in time.

We report each model's GPQA score as a common reference measure of capability; random-choice performance on GPQA is 0.25. The three highest-scoring models within each period are underlined. Closed-source models are shown in italics. For models with explicit reasoning-effort controls, we use the provider's default reasoning level, or "medium" where an explicit setting is required. Missing scores correspond to evaluations that remain to be collected.

C Protocols Implementation

This section goes in detail into the implementation of each of the protocols we propose

C.1 Pairwise Judgements

[TODO: add very detailed working of PJ and diagrams]

C.2 Ensemble

We additionally test an ensemble of pairwise judgements, aggregated through mean of judgements and measure it agianst Pairwise Judgement. This ensemble uses 3 different judges, usually the top-3 model of the previous generation.

D Additional Results

[TODO: intro text]

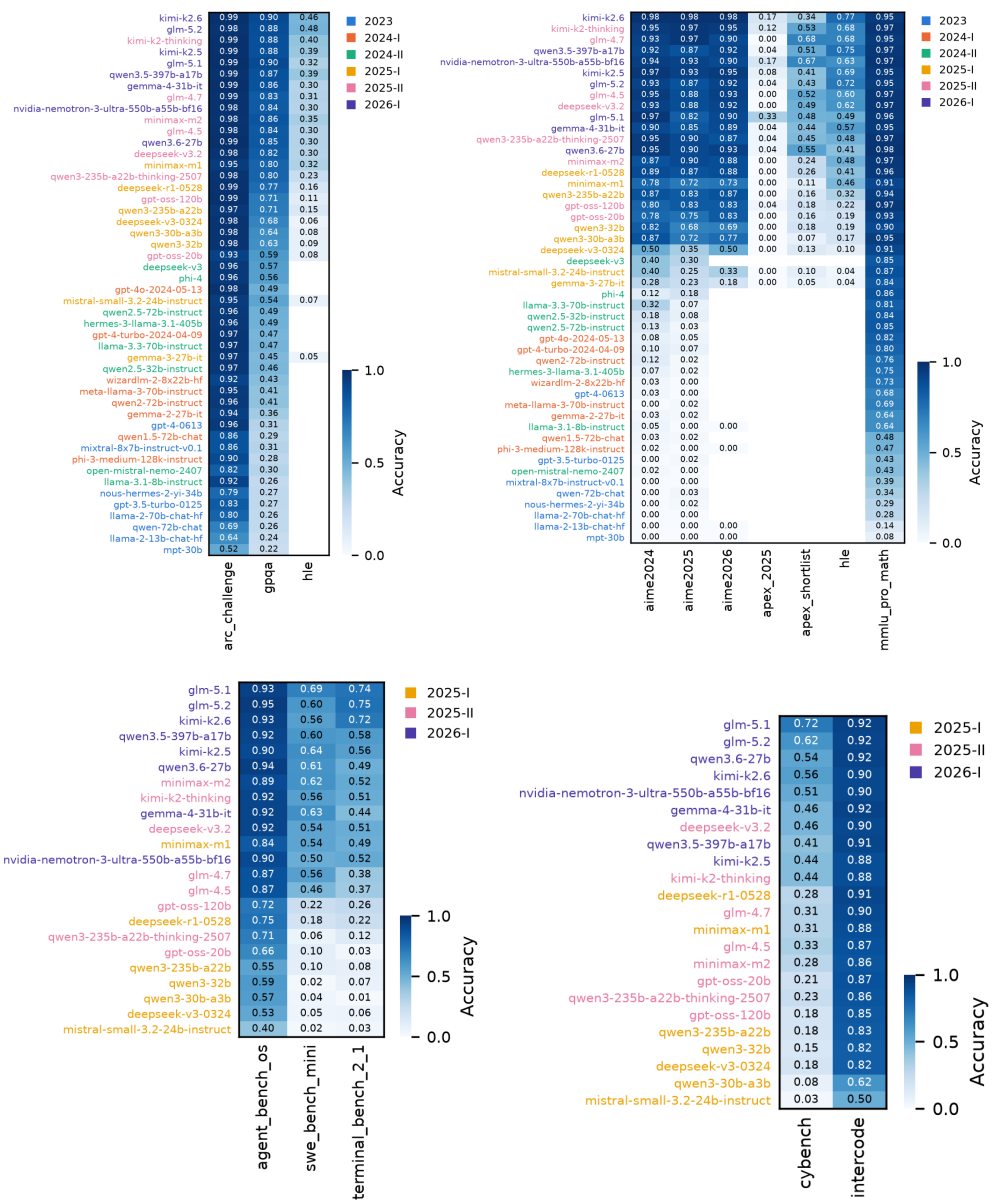


Figure 11: Accuracies for each model and collected benchmark.

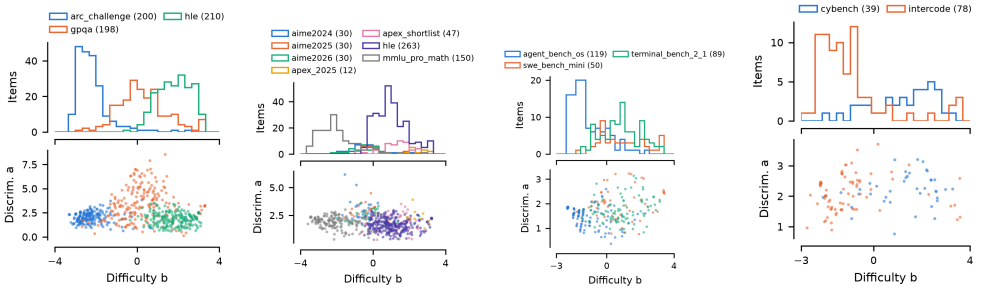


Figure 12: Difficulty scores b_i and a_i for each collected benchmark.

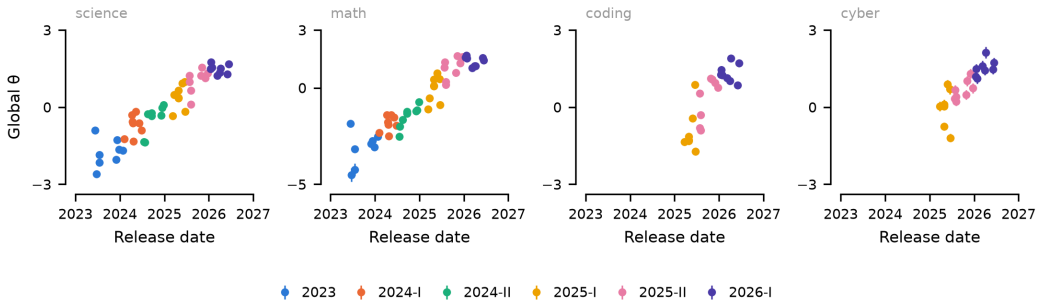


Figure 13: Computed strength value θ_m for each collected model across domains.

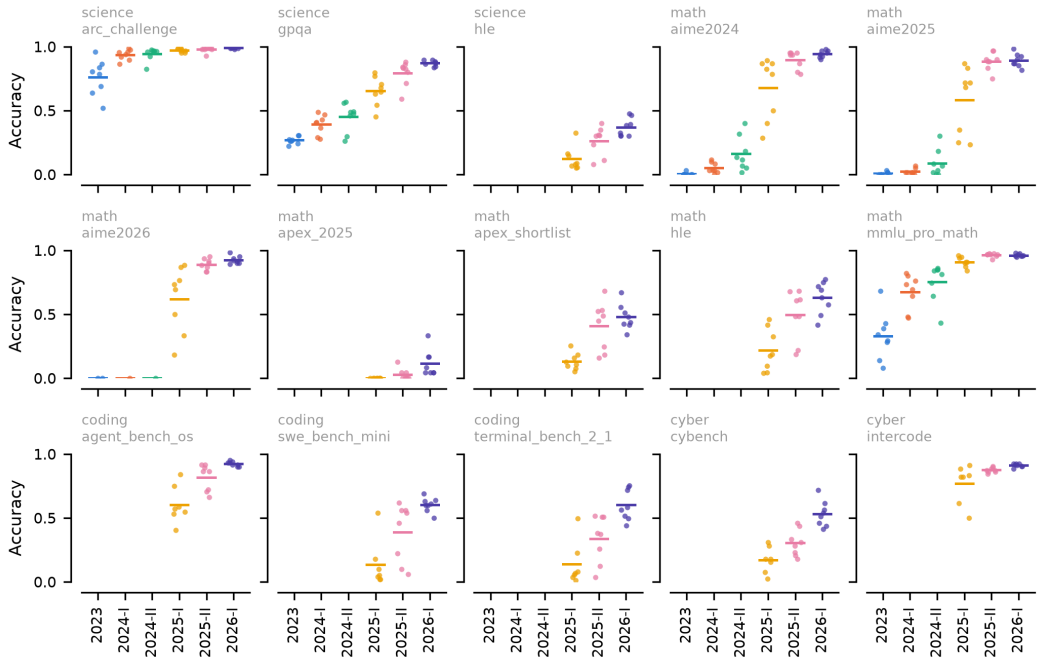


Figure 14: Score on the collected benchmarks by model and release date.

Table 3: **Models used in the retrospective evaluation.** Models are grouped by evaluation period. The three highest GPQA scores within each period are underlined, and closed-source models are shown in italics. GPQA has a random-choice baseline of 0.25. A dash indicates that the corresponding evaluation has not yet been collected.

Period	Model	GPQA
2023	*gpt-4-0613*	0.307
	mixtral-8x7b-instruct-v0.1	0.306
	llama-2-70b-chat-hf	0.263
	llama-2-13b-chat-hf	0.240
	mpt-30b	0.222
	qwen-72b-chat	–
	nous-hermes-2-yi-34b	–
	gpt-3.5-turbo-0125	–
2024-I	*gpt-4o-2024-05-13*	0.489
	gpt-4-turbo-2024-04-09	0.466
	wizardlm-2-8x22b	0.434
	qwen2-72b-instruct	0.408
	meta-llama-3-70b-instruct	0.406
	gemma-2-27b-it	0.365
	phi-3-medium-128k-instruct	0.276
	qwen1.5-72b-chat	–
2024-II	phi-4	0.561
	qwen2.5-72b-instruct	0.491
	llama-3.3-70b-instruct	0.474
	qwen2.5-32b-instruct	0.461
	open-mistral-nemo-2407	0.299
	llama-3.1-8b-instruct	0.259
	deepseek-v3	–
	hermes-3-llama-3.1-405b	–
2025-I	deepseek-r1-0528	0.769
	qwen3-235b-a22b	0.707
	deepseek-v3-0324	0.676
	qwen3-32b	0.631
	gemma-3-27b-it	0.439
	qwen3-30b-a3b	–
	minimax-m1	–
	mistral-small-3.2-24b-instruct	–
2025-II	kimi-k2-thinking	0.879
	glm-4.5	0.841
	glm-4.7	0.835
	deepseek-v3.2	0.824
	qwen3-235b-a22b-thinking-2507	0.801
	gpt-oss-120b	0.715
	gpt-oss-20b	0.588
	minimax-m2	–
2026-I	glm-5.1	0.899
	kimi-k2.6	0.897
	glm-5.2	0.884
	kimi-k2.5	0.876
	gemma-4-31b-it	0.865
	qwen3.5-397b-a17b	0.859
	qwen3.6-27b	0.859
	nvidia-nemotron-3-ultra-550b-a55b-bf16	0.837

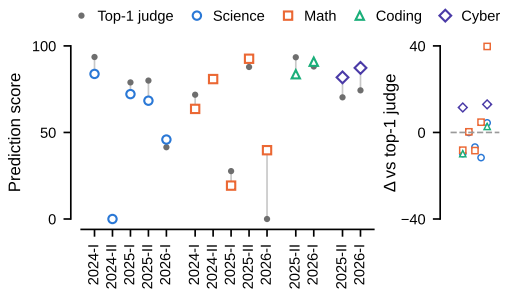


Figure 15: Performance of an ensemble of judges. In general performances don't particularly change, but for one specific bad judge the ensemble helps avoiding drastic error.

Pairwise judge prediction gain is concentrated in agentic domains

PLACEHOLDER FIGURE

Non-agentic vs. agentic tasks.
Optionally: agentic full trace vs. trace without tool calls.

Figure 16: **Capability prediction in agentic and non-agentic settings.** Prediction performance for tasks solved through a single model response and tasks that require interaction with tools or an external environment. Where available, we additionally compare full agentic traces against traces with tool interactions removed.

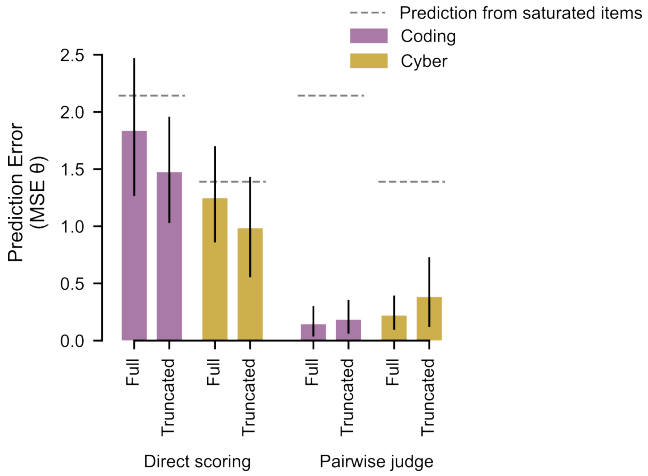


Figure 17: **Effect of transcript truncation.** Prediction error when LLM-based protocols receive the complete transcript or truncated versions containing less of the original response. This experiment approximates the effect of a judge whose context window is too small to observe the full model trajectory. [TODO: Specify truncation fractions or token budgets.]

Agentic and non-agentic settings. [TODO: may have to move this to appendix] Some of our domains require models to interact with an environment rather than produce a single static response. In these agentic settings, the transcript can contain tool calls in addition to natural-language reasoning, exposing different signals from those available in non-agentic question answering.

We therefore compare prediction performance between non-agentic tasks and agentic tasks, and separately test the contribution of tool-interaction traces when possible.

[TODO: analyze]

Limited context windows constrain access to transcript signal. [TODO: may have to move this to appendix]

A practical limitation of LLM-based judging is that model transcripts can be longer than the context window available to the judge. This is especially relevant for reasoning and agentic models, which may generate long intermediate traces. We therefore test how prediction performance changes when the transcript is progressively truncated before being given to the judging protocol.

[TODO: analyze]

Scaling with the amount of reference data. [TODO: probably move to appendix] Transcript-based prediction also requires reference models whose capability is already known. We therefore test how performance changes as we reduce either the number of available reference models or the number of transcripts observed for each model. This measures how much calibration data is required before the transcript signal becomes useful.

D.1 Number of evaluated items

To assess the stability of the evaluated protocols as the number of collected items increases, we ablate the number of items given to each protocol. For pairwise scoring, additional items sometimes improve the prediction (2025-II, Science 2024-I), but in most cases they provide little benefit. In at least one case (Math 2026-I) performance drops and doesn't recover as more items are added, which we attribute to a judge bias that gets amplified as data accumulates.

For direct scoring the trend is more consistent, with prediction score increasing mildly as the number of judged items grows.

Overall this points to the number of items

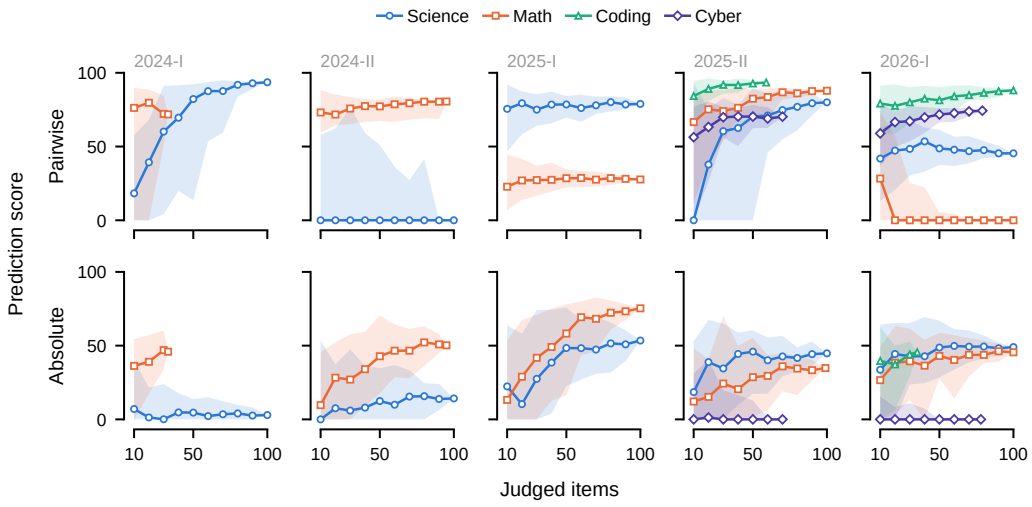


Figure 18: Changes in prediction score as we reduce the number of judged items.

D.2 Shifts in the reference population

To assess the stability of the pairwise protocol as the population of reference items change, we ablate this in two directions: (1) keeping only the stronger models of the reference cohort, (2) keeping olnly the weaker models of the reference cohort.

We note that the latter is way more destuctive than the former, showcasing that the presence of stronger reference models solidly increases the performance of the protocol as it enables anchoring.

Table 4: **Within-domain prediction performance.** Relative MSE when a protocol is calibrated and evaluated within the same capability domain, averaged across target periods. Values below 1 indicate lower prediction error than the benchmark-accuracy baseline.

Protocol	Science	Mathematics	Coding	Cybersecurity
Benchmark accuracy	1.000	1.000	1.000	1.000
Pairwise transcript comparison	**0.338**	**0.690**	**0.185**	**0.205**
Direct transcript scoring	1.366	1.488	1.900	1.480
Transcript length	0.842	1.006	2.410	2.930
Length + prose features	8.230	116.458	0.645	1.365
Transcript embedding	0.752	1.138	1.705	1.410

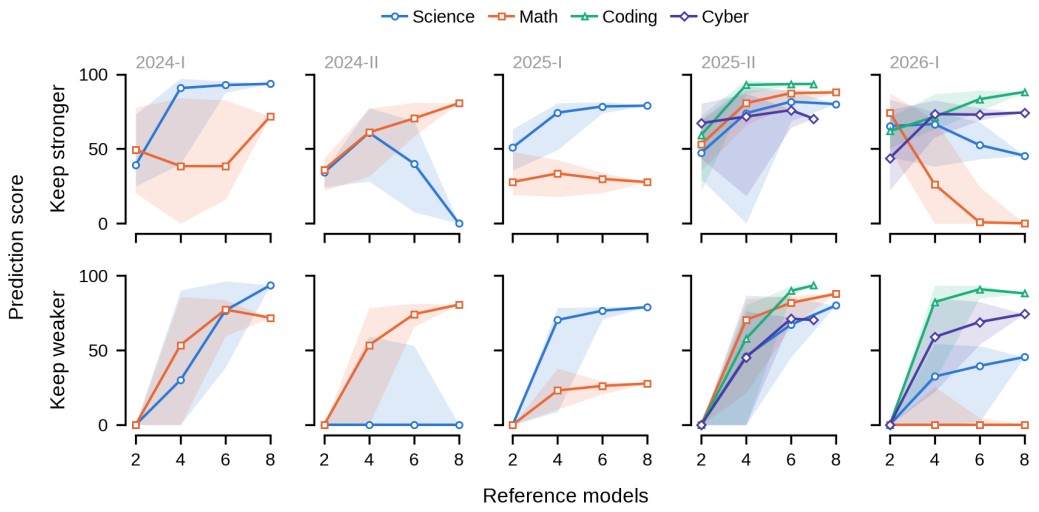


Figure 19: Changes in prediction score as the population of reference models change.

Within-domain prediction. We first evaluate each protocol within the same capability domain. Table 4 reports relative MSE separately for science, mathematics, coding, and cybersecurity, averaged across the available target periods within each domain.

Pairwise transcript comparison is the only tested protocol that consistently improves on benchmark accuracy across all four domains, with its relative MSE ranging from 0.185 in coding to 0.690 in mathematics. Simpler transcript-based approaches are considerably less stable. Full-transcript embeddings improve on the benchmark baseline in science but underperform it in mathematics, coding, and cybersecurity, while transcript length performs competitively only in science.